Pranav Kalakota

West Lafayette, IN | pkalakot@purdue.edu | linkedin.com/in/pranav-kalakota | github.com/Pranavkalakota

Education

Purdue University West Lafayette, IN

Bachelor of Science in Computer Science *Aug. 2025 – May 2028*

- Coursework: Programming in C, Object-Oriented Programming, Linear Algebra, Discrete Mathematics, Statistical Methods

Experience

Data Science Researcher Aug. 2026 – Present

Johnson & Johnson *West Lafayette, IN*

- Building a **Copilot Studio** agent over **Power Automate** cloud flows to automate JJ's **45-minute, 10-step booking process**, using **Microsoft Graph** for calendar availability and attendee resolution.
- Designing a **multi-agent Copilot Studio** architecture within an **8,000-character context window**, compressing instructions into Word doc references to reclaim **2,000+ characters** for scheduling logic.
- Sandboxing JJ's Microsoft environment across a **12-person team**, using **Power BI** to benchmark prompt reliability and reduce Copilot's **10% template-adherence failure rate**.

Software Engineering Intern June 2026 – Present

Crcle *West Lafayette, IN*

- Architected a **multimodal RAG pipeline** fusing **Whisper ASR**, **Llama 3.2 3B**, **LLaVA 7B**, **ArcFace**, and **ECAPA-TDNN** with cosine-similarity gating over **ChromaDB** ONNX vector store, enabling zero-touch HCI via cross-app voice control with **sub-20s latency** and zero cloud dependency.
- Engineered a schema-constrained **NLP intent router** with **3-tier fuzzy/regex/LLM fallback** and semantic vector search, achieving **95% deterministic routing** across **500+ fuzz invariants** and 12+ query types in production.
- Optimized always-on wearable ASR pipeline with adaptive windowing and CPU-routed **Whisper** inference, cutting transcription latency **50% (12s to 6s)** and wake-word activation **90% (2s to <200ms)** via offline-first **ONNX** detection with parallel ASR fallback.
- Built a cross-platform **vision-based GUI automation** system achieving **sub-2s latency** and **90%+ element localization accuracy** across arbitrary applications by routing screenshots through **Gemini** multimodal LLM inference with pixel-precise coordinate extraction.

Software Engineering Intern June 2026 – Aug. 2026

Pololu Robotics *Las Vegas, NV*

- Engineered an automated production test system in Arduino with serial CLI and fault handling, validating dual H-bridge MOSFET output, current thresholds, and fault flags across **15+ ADC channels**, cutting QA cycle time by **10%**.
- Refactored **500+ line** legacy embedded codebase, resolved a critical **timing race condition** in PWM chopping logic, and validated power-on-reset sequencing for long-term maintainability.

Projects

LSM-Tree Storage Engine | *Java, Maven, Guava* Aug. 2026 – Oct. 2026

- Built a **crash-safe LSM-tree engine** in **Java** sustaining **481K** buffered writes/sec and **2.46M** random reads/sec at **0.88μs p99** across **100K** keys.
- Amortized **fsync (3.8ms)** via **group commit** off the write lock: synced throughput **263→829/sec** at 8 writers, **3.2×** with durability unchanged.
- Cut reads to **one SSTable per level** via **leveled compaction**, reclaiming **84%** of disk; **LZ4** shrank repetitive blocks **87%** and lifted cold reads **968K→1,063K/sec**.

MCP Task Manager | *TypeScript, SQLite, Express, MCP Protocol* Jan. 2026 – Mar. 2026

- Engineered a dual-architecture task management system exposing **5 tools over MCP** via stdio transport with an **Express** web dashboard backed by a shared **SQLite** database.
- Designed a **graceful degradation layer** falling back from **OpenAI API** to a local regex-based parser with fuzzy typo tolerance, ensuring **zero downtime** during API failures.

Honors & Awards

Claude Builder Hackathon: 1st place in the healthcare track for VEDA
Databricks Generative AI Fundamentals
Deloitte Data Analytics Certificate

Technical Skills

Languages: Java, Python, C, TypeScript, Swift, HTML/CSS
Tools & Platforms: Maven, Arduino, ChromaDB, Copilot Studio, Power Automate, Power BI, Microsoft Graph
Libraries & Frameworks: Guava, Express, MCP, Whisper, Llama 3.2, LLaVA, ArcFace, ECAPA-TDNN
Concepts: Object-Oriented Programming, Embedded Systems, RAG Pipelines, LSM Trees, Write-Ahead Logging